

Enhancing Urban Traffic Sustainability: Anomaly Detection in Time Series Incorporating Spatial Correlations and Temporal Evolutions

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Abstract

The efficiency of Intelligent Transportation Systems (ITS) in smart cities hinges on the accurate identification of traffic anomalies, with traffic time series data serving as the primary data source. However, the ever-increasing amount of data poses challenges to traffic practitioners in effectively detecting or labeling anomalies, due to the prohibitive cost of manual calibration and the complex spatial and temporal dynamics of urban traffic. Addressing this challenge, this paper investigates the intrinsic spatial dependencies and temporal evolution within traffic time-series data, leading to the development of two independent, data-driven anomaly detection algorithms for spatial and temporal anomaly detection. The spatial anomaly detection algorithm employs the Maximal Information Coefficient approach to identify abnormal roads with unusual traffic patterns. The temporal anomaly detection algorithm leverages the mechanism of temporal evolution dynamics in the Markov Transition Field, catering to analytical needs across different time granularities. Validation is conducted on four real-world traffic speed datasets, and the results confirm the effectiveness of the proposed algorithms. By simplifying data requirements and harnessing spatial-temporal insights, the proposed anomaly detection algorithms can identify spatial and temporal anomalies with just time-series data, representing a step towards more efficient and adaptable traffic management systems for practical use.

Keywords: Anomaly Detection, Data-driven, Intelligent Transportation System (ITS), Time Series

1. Introduction

Intelligent Transportation Systems (ITS) are crucial components of smart cities, aiding in providing effective traffic management, enhancing mobility, and reducing energy costs and environmental pollution (Rathore et al., 2021; Salman and Hasar, 2023). However, the effectiveness of ITS is often hampered by abnormal traffic patterns, including traffic congestion, accidents, special events, or atypical traffic on specific roads (Parkany and Xie, 2005; Zhang et al., 2016). Traffic anomalies may result in stop-and-go traffic and heightened congestion within urban networks (Qian et al., 2020), consequently impairing smart city operations and impeding the development of a green and sustainable system. Detecting and understanding these abnormal traffic patterns could help develop effective traffic management systems and control policies, thereby reducing congestion duration, cutting emissions, and improving traffic safety (Zeroual et al., 2019; Zhou et al., 2021).

During the current development stage of the ITS, traffic time-series data from sensors serves as the fundamental dataset for further management (Ahmed et al., 2022). Within this framework, time-series data analysis is instrumental in identifying both spatial and temporal anomalies. Considering speed time-series data, generally, most road segments demonstrate some degree of deterministic characteristics (Vlahogianni et al., 2008) owing to their periodic daily traffic patterns, and traffic patterns on adjacent road segments often show notable similarities (Anbaroglu et al., 2014; Zhang et al., 2016). However, certain road segments may exhibit more irregular or stochastic patterns, as observed in their overall time-series data, leading to a diminished level of spatial dependence with adjacent road segments. Such segments, characterized by their reduced correlation or interaction with neighboring roads in terms of traffic conditions, can be identified as *spatial anomalies*. Furthermore, analyzing the time-series data for each road segment at various

time granularities—ranging from minutes to days—reveals *temporal anomalies*. The temporal anomalies are characterized by either rapid fluctuations or exceedingly low-speed values in the specific time intervals of road segments, and can be attributed to factors such as traffic congestion, accidents, special events, or holidays.

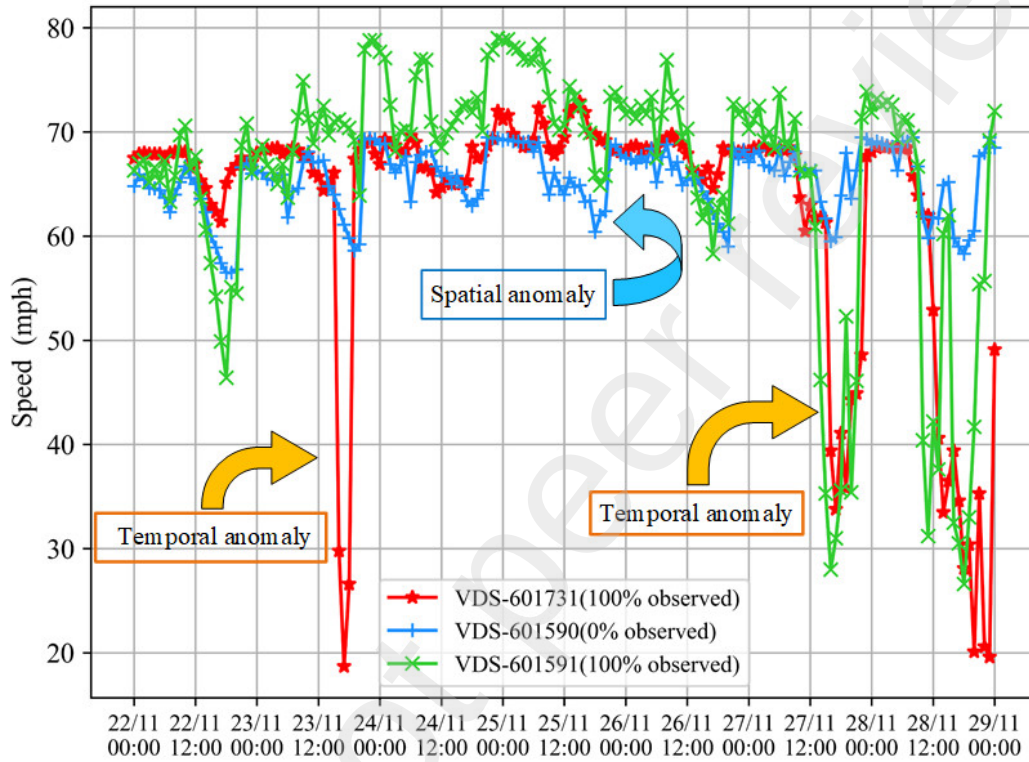


Fig. 1. Illustration of spatial and temporal anomalies in time-series traffic speed data

To visually demonstrate spatial and temporal anomalies in speed data, we present traffic speed data collected from three vehicle-detector-stations (VDS) within the California Department of Transportation (Caltrans) Performance Measurement System (PeMS)¹. Fig.1 showcases one week of traffic speed data, with a sampling granularity of one hour, obtained from three stations (VDS ID 601590, 601591, and 601731, arranged from upstream to downstream). The data for each of the three stations is depicted respectively in blue, green, and red. Both VDS-601591 and VDS-601731 displayed relatively low values during the weekends (27th and 28th November 2021), which

¹ <http://pems.dot.ca.gov>

could be classified as temporal anomalies. Additionally, speed drops observed at VDS-601731 on 23rd November could also be considered as a form of temporal anomaly. Regarding spatial anomalies, one of the three stations, VDS-601590, displayed a different speed pattern from the other two nearby stations. This discrepancy can be attributed to the poor data quality of VDS-601590, where 0% of vehicles were observed, as reported in the data quality report from PeMS. In comparison to its upstream and downstream counterparts, this abnormal station deviated from the expected traffic pattern and exhibited less deterministic characteristics.

A basic assumption in anomaly detection is that the majority of traffic time-series data follow similar patterns, while anomalies represent a minor proportion of the data (Ahmed et al., 2022; He et al., 2023). Various efforts have been developed to identify abnormal traffic patterns, including change point detection algorithms, traffic flow theory models, and prediction-based methods (Alesiani et al., 2018). However, two challenges remain to be addressed as follows:

(i) How can anomalies be detected in limited data environments, particularly with single-source traffic time-series data? The limited data environment refers to the current situation where ITS in most cities can only provide limited traffic data. While accessing and analyzing high-dimensional multi-source traffic data can offer system-wide efficiency and objective evaluations (Yang et al., 2021), it is worth noting that the availability of such data in practical scenarios is often limited or insufficient. In most cases, traffic data accessible through ITS are limited to basic meta-information, such as time stamps, road or loop IDs, and some speed/flow/density combinations, while only a few regions with well-developed ITS can provide more detailed traffic data. For example, video surveillance systems that can greatly help in traffic anomaly detection are only equipped on a few freeways (Alesiani et al., 2018).

Therefore, in practical scenarios, traffic data used for detecting anomalies may be limited,

where time-series data such as speed, volume, and occupancy are still the most commonly used data sources. In such a scenario, developing data-driven detection methods that can be directly applied to single-source traffic time-series data becomes particularly necessary. These methods need to effectively handle limited data resources and accurately identify traffic anomalies without requiring additional information support.

(ii) How might one explore the spatial correlations and temporal evolutionary characteristics in anomaly detection? Accurate spatial-temporal representation is important in comprehensively depicting traffic dynamics and integrating ITS applications (Vlahogianni et al., 2014). However, the exploration of spatial correlations and temporal evolutions within historical traffic data remains insufficient. Despite extensive attempts to detect anomalies in data streams based on immediate past traffic data, the wealth of historical traffic information continues to be largely underutilized (Chakraborty et al., 2019). Simply extracting features at a specific time or road segment is insufficient to fully capture the stochastic nature of traffic flow evolutions (Yang et al., 2021) and spatial dependencies (Zhang et al., 2022). Moreover, while supervised learning methods (Yuan et al., 2019; Zhang et al., 2018) can capture non-linear correlations to detect anomalies, the efficacy of these learning-based approaches is constrained by the scarcity of ground-truth anomaly labels in traffic time-series data. This limitation primarily stems from the significant cost of manual calibration (He et al., 2023). Consequently, the necessity of developing a data-driven approach becomes apparent, one that utilizes the data's intrinsic patterns and characteristics to identify anomalies within time-series data through unsupervised methods, thereby obviating the need for exact anomaly labels.

Motivated by existing research gaps in traffic anomaly detection, this study introduces a hybrid data-driven approach to address spatial and temporal anomaly detection tasks in traffic

time-series data. This approach leverages two distinct methods to explore the spatial correlations and temporal evolutionary characteristics inherent in single-source time-series data. In the spatial domain, the Maximal Information Coefficient (MIC) ([Reshef et al., 2011](#); [Reshef et al., 2016](#)) is employed to capture complex linear and non-linear spatial dependencies in time-series data across roads. The MIC is capable of quantifying the mutual information ([Cover, 1999](#)) between time series using an adaptive grid partition process ([Reshef et al., 2011](#)), thereby measuring a wide range of associations stemming from real-world interactions. Regarding the temporal domain, the Markov Transition Field (MTF) method ([Wang and Oates, 2015](#)) is used to depict the temporal evolution characteristics within traffic time-series data, modeling the segment states of each time-series sequence as transitioning probabilities over time. The MTF method effectively captures the temporal dynamics and dependencies within the data, facilitating analysis and understanding how the traffic patterns evolve.

By combining the spatial insights provided by the MIC with the temporal analysis capabilities of the MTF method, this study achieves a comprehensive understanding of both spatial and temporal dimensions in traffic time-series data. This hybrid anomaly detection approach not only identifies spatial anomalies among roads but also captures temporal anomalies within traffic evolution patterns. The main contributions of this work can be summarized as follows:

- We establish a MIC-based spatial anomaly detection algorithm that provides a solution for identifying abnormal roads. The algorithm can accurately capture the complex linear and non-linear spatial dependencies among roads and identify abnormal roads with more stochastic patterns, which has not been adequately addressed in clustering-based studies.
- Combined with the MTF method, our proposed temporal anomaly detection algorithm capitalizes on the inherent temporal evolution characteristics within traffic time-series

data. The algorithm can effectively detect temporal anomalies across various granularities by quantifying the self-transition probabilities of temporal segment states. This method is well-suited for limited data environments relying on single-source information (utilizing only time-series data), independent of prior knowledge or external data sources.

- The detection results of the proposed method enable the identification of roads with more chaotic traffic patterns requiring additional monitoring for traffic management and control, as well as the labeling of temporal anomaly points within a large amount of time-series traffic data, all without necessitating manual calibration.

The rest is organized as follows: Section 2 reviews current studies on urban road traffic anomaly detection; Section 3 explains the preliminaries and mathematical formulation of the spatial-temporal anomalies detection method; Section 4 presents the detection results and pattern analyses through real-world traffic speed data; Section 5 summarizes the conclusions and potential future works.

2. Literature Review

2.1 Spatial Anomaly Detection

Spatial anomaly detection in existing studies can be broadly classified into two categories: abnormal trajectory detection and abnormal traffic pattern detection. Abnormal trajectory detection has been extensively studied in the context of airspace operations, where clustering techniques like Principal Components Analysis (PCA) (Gariel et al., 2011), Gaussian Mixture Model (GMM) (Li et al., 2016), and Density-Based methods (Deshmukh and Hwang, 2019; Wang et al., 2019) have been applied to identify abnormal data. In the domain of road traffic, anomaly detection focuses on GPS trajectories and aims to identify abnormal trajectories that deviate from the norm (Ahmed et al., 2022). Furthermore, data fusion methods with other data sources, like loop data (Canepa and

[Clausen, 2013](#)), have been employed to improve anomaly detection accuracy.

In terms of abnormal traffic pattern detection, studies can be classified into image-based anomaly detection and time-series anomaly detection ([Javed et al., 2020](#)). This paper primarily focuses on detecting spatial anomalies from time series. The underlying assumption for detecting spatial anomalies is that adjacent roads exhibit a certain degree of spatial correlation due to traffic propagations. In general, detecting these types of spatial anomalies can be achieved by clustering methods that employ various metrics such as density, flow and adjacent matrix, or secondary developed measurements such as entropy and frequency of occurrences ([Anbaroglu et al., 2014](#); [Zhang et al., 2016](#)). However, investigating complex intrinsic correlations among roads with limited data information remains a challenge when exploring spatial anomalies.

2.2 Temporal Anomaly Detection

As for the temporal domain, various methods have been proposed to detect temporal anomalies, which are often associated with change point and event detection techniques ([Gupta et al., 2013](#)). Cumulative sum algorithm (CUSUM) ([Teng and Qi, 2003](#)), together with California algorithms and their variants ([Petty et al., 2002](#); [Stephane De S and Chassiakos, 1993](#)), were used to identify different traffic patterns by pair-wise comparison between upstream and downstream sensors. Besides, some studies tended to set a threshold based on historical data to distinguish normal and abnormal traffic conditions. The most representative methods for this purpose were standard normal deviation (SND) and its derivations ([Chakraborty et al., 2019](#); [Snelder et al., 2013](#); [Tang and Gao, 2005](#)). In order to prevent interferences from outliers, the SND family of methods required the exclusion of accurate start and end times of traffic incidents when calculating the threshold. In addition, hyperparameter tuning was necessary for these methods, adding complexity to the process. Consequently, the SND methods required stringent data requirements and

adjustments.

Although these studies achieved satisfactory performances, some claimed there was still room for improvement, e.g., considering traffic impacts by combining traffic flow theories. For instance, Asakura et al. (Asakura et al., 2017) employed shock wave theory in the GPS trajectory of probe vehicles to detect traffic incidents. Other studies utilized the Lighthill–Whitham–Richards (LWR) model (Canepa and Claudel, 2013; Zeroual et al., 2019; Zeroual et al., 2017) and fundamental diagrams (Kalair and Connaughton, 2021) as detection aids. However, these methods that incorporated traffic flow theories often required a robust database with multiple types of traffic data (such as flow and density) to construct accurate fundamental diagrams. Moreover, implementing shock wave analysis at a network scale can be costly and challenging.

In recent years, data mining methods such as machine learning and deep learning have gained popularity due to the availability of large amounts of data. Chen et al. (Chen et al., 2016) proposed a systematic approach by secondary incident identification and K-Nearest Neighbor (KNN) pattern matching to analyze non-recurrent congestions. Mercader and Haddad (Mercader and Haddad, 2020) applied the isolation forest to detect isolated anomalies in traffic data from Bluetooth sensors. Support Vector Machine (SVM) has also been used for incident detection, as demonstrated in (Wang et al., 2013). Additionally, several studies took advantage of deep learning methods and proposed incident detection methods based on Deep Belief Network (DBN) (Zhang et al., 2018), Long Short-Term Memory (LSTM) (Yuan et al., 2019), Convolutional Neural Network (CNN) (Cai et al., 2020), and Autoencoder (Islam et al., 2021).

Despite the progress in traffic speed data collection and computation (Mercader and Haddad, 2020), the effectiveness of learning-based models in anomaly detection relies heavily on the quality of input data, particularly accurate anomaly labels. However, obtaining precise anomaly

labels is challenging, and their availability is often limited, even when the dataset is extensive. Manual efforts are typically required to label abnormal instances accurately, which is time-consuming and expensive. Moreover, it is crucial to exclude abnormal data points from the training data to ensure the development of a reliable model that represents normal traffic patterns accurately (Chakraborty et al., 2019). Manually improving the quality of collected data is impractical, especially for large-scale urban networks. Therefore, there is a pressing need for data-driven algorithms that leverage the time-series characteristics of traffic data to label temporal anomalies automatically. Such algorithms can significantly reduce the dependence on external labels and enable the development of more effective and scalable anomaly detection models.

3. Method

3.1 Preliminaries

This subsection introduces the foundational concepts and overall structure of the proposed method. The time-series data used in this study consists of multivariate spatial-temporal information, including traffic speed data for each road segment across a specified timeframe. Mathematically, considering a discrete-time period of T ($T \in N^*$), the traffic speed data of S ($S \in N^*$) road sections can be regarded as a matrix $\mathbf{V}_{T \times S}$.

$$\mathbf{V}_{T \times S} = \begin{pmatrix} V_1^1 & \dots & V_1^S \\ \dots & \dots & \dots \\ V_T^1 & \dots & V_T^S \end{pmatrix}_{T \times S} \quad (1)$$

Formally, the spatial and temporal anomalies in this paper are defined as follows:

(i) Spatial anomalies refer to road sections with traffic patterns that diverge from the majority within the urban network. While traffic propagation typically leads to similar patterns among adjacent roads, spatial anomalies often feature time-series patterns with much more stochastic characteristics than typical roads. (Anbaroglu et al., 2014; Zhang et al., 2016). In matrix $\mathbf{V}_{T \times S}$, the

spatial anomalies are identified as abnormal columns.

(ii) Temporal anomalies in this paper refer to rapid changes or extremely low-speed values in time-series data at different levels of temporal granularity, such as every few minutes and every day. The occurrence of low-speed values may indicate traffic congestion or accidents, which could result in a decrease in traffic service level. In matrix $V_{T \times S}$, the temporal anomalies are regarded as abnormal points in each column. An anomaly score will be assigned to each temporal stamp, and a larger score will indicate anomalies (Kandanaarachchi, 2022).

Moving from the foundational definitions, we now introduce the overall framework structure of our hybrid data-driven spatial and temporal anomaly detection method, depicted in Fig. 2, laying the groundwork for the subsequent mathematical details.

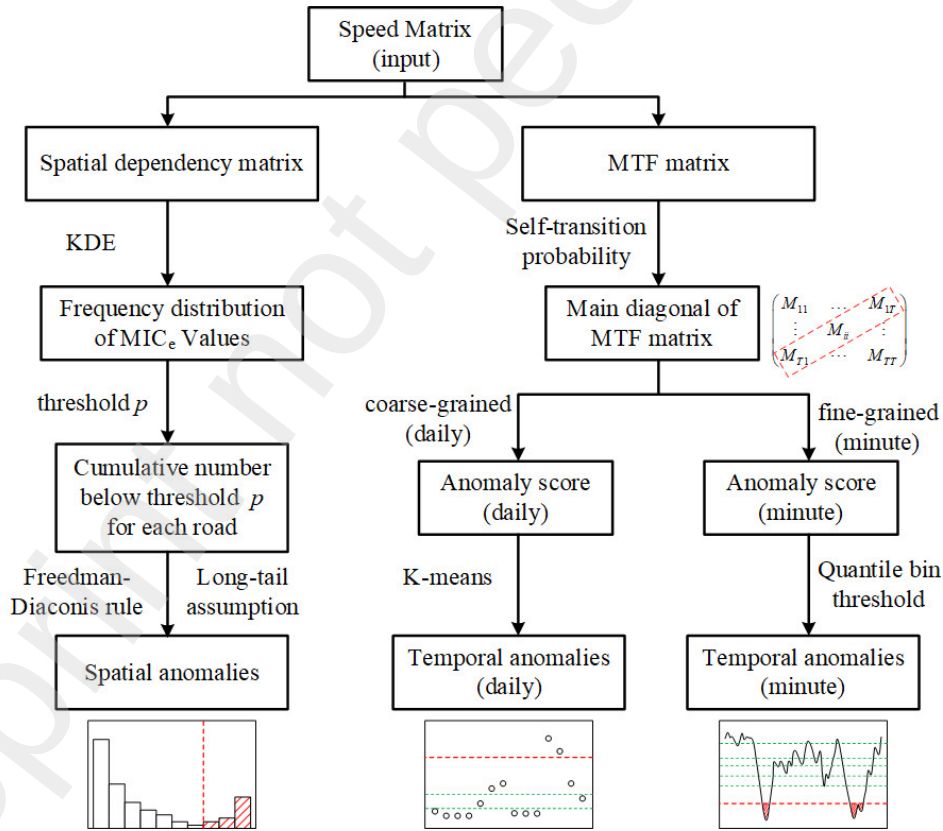


Fig. 2. Hybrid data-driven spatial and temporal anomaly detection method

The framework outlines the hybrid data-driven approach for detecting anomalies in traffic

speed time-series data, encompassing both spatial and temporal domains. Starting with the speed matrix as input, spatial dependencies between roads are computed through the MIC-based method and represented in a matrix form. Kernel Density Estimation (KDE) (Chen, 2017) is then utilized to establish a frequency distribution, with a subsequent threshold set to identify weak spatial dependencies. For each road segment, the cumulative instances of spatial dependence falling below the threshold with other segments are calculated. A histogram of these cumulative values is constructed using the Freedman-Diaconis rule (Freedman and Diaconis, 1981). Road segments exhibiting anomalous behavior are then identified through the premise that spatial dependencies follow a long-tail distribution. In parallel, the speed matrix is transformed into the MTF matrix, focusing on self-transition probabilities along the main diagonal to reflect the temporal stability of traffic states. Anomaly scores for both daily and minute granularity are calculated, with daily anomalies categorized by K-means clustering and minute-level anomalies identified using a topside quantile bin threshold. This hybrid approach is adept at identifying spatial anomalies, which are presented in histograms, as well as temporal anomalies, depicted through point plot graphs for both daily and minute intervals. This facilitates a thorough analysis of traffic anomalies from both qualitative and quantitative perspectives. Details of the method are discussed in subsequent sections.

3.2 MIC-based Spatial Anomaly Detection Algorithm

The objective of detecting spatial anomalies in the multivariate speed matrix $\mathbf{V}_{T \times S}$ is to identify road sections that exhibit weak inner spatial dependences with others, as such roads violate the traffic propagation law or display much more stochastic than ordinary roads. However, it is challenging to capture certain non-linear spatial dependencies in multivariate traffic data through some commonly used linear-based correlation measurements such as Spearman and Person correlation coefficient (Dai

et al., 2019; Habtemichael and Cetin, 2016). To overcome this challenge, we employed the MIC (Reshef et al., 2011; Reshef et al., 2016) to mine both linear and non-linear relationships in large-scale time-series traffic data, as follows:

$$MIC(X;Y) = \max_{ab \leq B(n)} \frac{I^*((X;Y), a, b)}{\log_2 \min(a, b)} \quad (2)$$

where, $MIC(X;Y)$ is the MIC value between variable sets X and Y ; $\log_2 \min(a, b)$ is used for normalization. n is the number of road sections and $B(n) = n^{0.6}$, according to (Reshef et al., 2016). $I^*((X;Y), a, b)$ is the maximum value of mutual information (Cover, 1999) obtained by the grid partition with a rows and b columns on variable sets X and Y . Note that the calculation formula of mutual information is omitted here because it can be induced by grid partition, where the value is proportional to the number of data points falling inside the box (Reshef et al., 2011).

Basically, the MIC can encapsulate the linear or non-linear relationships between two variable sets by partitioning a grid of the scatterplot of the two sets (Reshef et al., 2011). To speed up the calculation, we employ a consistent estimator called MIC_e (Reshef et al., 2016) by simplifying the grid partition process. The MIC_e is defined as follows:

$$MIC_e(X;Y) = \begin{cases} \max_{ab \leq B(n)} \frac{I_e^*((X;Y), a, [b])}{\log_2 \min(a, b)} & \text{if } b \geq a \\ \max_{ab \leq B(n)} \frac{I_e^*((X;Y), [a], b)}{\log_2 \min(a, b)} & \text{otherwise,} \end{cases} \quad (3)$$

where $I_e^*((X;Y), a, [b]) = \max_{G \in G(a, [b])} I((X;Y)|_G)$; $G(a, [b])$ represents the grid partition with a rows and b columns on variable sets X and Y ; Note that $[b]$ denotes the partition condition that for $b > a$, the a -by- b grids are equipartition of size b in column and analogous to the $a > b$ condition (Reshef et al., 2016). The calculation results of MIC_e are normalized within the range of $[0,1]$, where a value of 1 indicates a strong correlation between the two variables, while a value of 0

indicates independence between them.

Given the multivariate speed matrix $\mathbf{V}_{T \times S}$, Eq. (3) calculates the spatial dependence between any pair of column sets (road sections) as $MIC_e(\mathbf{V}_{T \times S}; \mathbf{V}_{T \times S})$. This approach captures both inner linear or non-linear spatial dependencies among road sections. The resulting spatial dependency matrix $\mathbf{SD}_{S \times S}$ can be obtained as shown in Eq.(4).

$$\mathbf{SD}_{S \times S} = \begin{pmatrix} mic_e^{1,1} & \cdots & mic_e^{S,1} \\ \cdots & mic_e^{i,j} & \cdots \\ mic_e^{1,S} & \cdots & mic_e^{S,S} \end{pmatrix}_{S \times S} \quad (4)$$

where the element $mic_e^{i,j}$ ($mic_e^{i,j} \in [0,1]$) denotes the spatial dependence value between road i and road j . A higher value of $mic_e^{i,j}$ indicates a stronger relationship between road i and road j is. Notably, the spatial dependence value in the leading diagonal, which represents the auto-correlation of the road, is set to the maximum value of 1. It can be easily inferred that the spatial dependency matrix $\mathbf{SD}_{S \times S}$ is symmetric and $mic_e^{i,j} = mic_e^{j,i}$.

Based on the spatial dependency matrix, the spatial anomaly detection algorithm can be defined as follows (Algorithm 1).

Algorithm 1 Spatial anomaly detection algorithm

Input: Spatial dependency matrix $\mathbf{SD}_{S \times S}$.

Output: Cumulative number of MIC_e values $\mathbf{cn} = \{cn_1, \dots, cn_s\}$; anomaly road sections set

$\mathbf{AR} = \{ar_1, \dots, ar_c\}$ where c is the total number of anomaly road sections.

- 1: **for** each column j in $\mathbf{SD}_{S \times S}$ **do**
 - 2: Compute frequency distribution of MIC_e values in each column (kernel density estimation is used);
 - 3: **end for**
 - 4: Determine a low threshold $p(p \in [0,1])$ through the frequency distribution result and set $Flag = 0$;
 - 5: **for** each column j in $\mathbf{SD}_{S \times S}$ **do**
 - 6: **for** each row i in column j **do**
 - 7: **if** $mic_e^{i,j} \leq p$ **then**
 - 8: $Flag = Flag + 1$;
-

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9:      end if
10:   end for
11:   Let  $cn_j = Flag$ ,  $Flag = 0$ ;
12: end for
13: According to FD rules, compute the frequency distribution of elements in  $\mathbf{cn} = \{cn_1, \dots, cn_S\}$ ;
14: Set anomalies divided by FD rules as  $\mathbf{AR} = \{ar_1, \dots, ar_c\}$ ;

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Firstly, the frequency distribution of MIC_e values of each column (road) is computed according to kernel density estimation to determine a threshold p of MIC_e values (as shown in lines 1~4 in Algorithm 1). The value of p can be determined according to the distribution of MIC_e values. If the MIC_e value of a pair of roads is smaller than p , these two roads are considered as less correlated. The cumulative number of MIC_e values below the threshold p is counted for each road, and the values are stored in a set $\mathbf{cn} = \{cn_1, \dots, cn_x, \dots, cn_S\}$, where cn_x denotes the cumulative number of MIC_e below the threshold of road section x ($x \in S$). This procedure refers to lines 5~12.

Now each road section has its cumulative number, which can be used to determine anomalies. A large cumulative number indicates that few roads have strong correlations with the target road, and this target road may be an abnormal one. Subsequently, the frequency distribution of elements in the cumulative number set $\mathbf{cn}$ is computed using the Freedman-Diaconis rule (FD) (Freedman and Diaconis, 1981). The FD is designed to select the width of the bins in a histogram, and based on the FD result, the algorithm identifies the outlier part of the distribution that can be treated as an anomaly set of roads (i.e., lines 13-14 in Algorithm 1). Note that the FD estimator is robust (resilient to outliers) for long-tail distributions, where anomalies are typically located. By implementing Algorithm 1, the anomalous road sections $\mathbf{AR} = \{ar_1, \dots, ar_c\}$ can be deciphered, where c is the total number of anomalous road sections.

3.3 MTF-based Temporal anomaly detection algorithm

In this study, temporal anomalies are defined as rapid changes or extremely low values of traffic speed, and may include accidents, road constructions, public events, and holidays. To capture the inherent temporal dependencies present within each column (representing individual roads) of the traffic speed matrix $\mathbf{V}_{T \times S}$, this study introduces a temporal anomaly detection method utilizing the Markov Transition Field (MTF) approach (Wang and Oates, 2015). This method models traffic time-series features as a first-order Markov chain, preserving sequential characteristics within the temporal domain. It allows for measuring transition probabilities within time-series traffic data, thereby enabling the identification of temporal anomalies. The specific details of the temporal anomaly detection method utilizing the MTF approach are outlined as follows:

Given a time-series (i.e., speed data of road x) $(V_1^x, \dots, V_T^x)$, the domain of values is divided into Q quantile bins, where each bin has the same number of points. Thus a $Q \times Q$ Markov matrix can be built by computing the transitions among bins in the manner of a first-order Markov chain (Wang and Oates, 2015). The quantile bin q_i ($q \in [1, Q]$) represents the bin that contains data of bin q at time i . Then the MTF is derived by extending the Markov matrix along with the temporal positions as Eq. (5).

$$\mathbf{M} = \begin{pmatrix} M_{11} & \dots & M_{1T} \\ \vdots & M_{ii} & \vdots \\ M_{T1} & \dots & M_{TT} \end{pmatrix} = \begin{pmatrix} w_{ij|v_1^x \in q_i, v_1^x \in q_j} & \dots & w_{ij|v_1^x \in q_i, v_T^x \in q_j} \\ \vdots & \ddots & \vdots \\ w_{ij|v_T^x \in q_i, v_1^x \in q_j} & \dots & w_{ij|v_T^x \in q_i, v_T^x \in q_j} \end{pmatrix} \quad (5)$$

where M_{ij} is the transition probability of bin q_i transferring to bin q_j (that is, $q_i \rightarrow q_j$), and w_{ij} is the frequency of a point in q_j followed by the other point in q_i ; Through this matrix, we can obtain transition probabilities from one timestamp to another and identify patterns of temporal

anomalies.

Note that the main diagonal of MTF, M_{ii} , corresponds to the self-transition probability of each bin, indicating the probability of maintaining the current traffic state in the next sample time period. A high self-transition probability implies that the traffic state is more likely to remain the same rather than transfer to other states. Generally, in situations where the traffic speed is extremely low, it is less likely for the traffic state to change, and hence, the timestamps with high self-transition probabilities can be identified as temporal anomalies.

In contrast to the original purpose of MTF for visualizing time series as a graph, our proposed algorithm leverages the transition dynamics embedded in MTF to detect temporal anomalies at multi-granularity. Specifically, in this study, we evaluate two temporal granularities, namely, 10 minutes (the minimum sample period in our dataset) and one day (144 sample periods). The multi-granularity temporal anomaly detection algorithm is presented in Algorithm 2.

Algorithm 2 Temporal anomaly detection algorithm

Input: Multivariate traffic speed matrix $\mathbf{V}_{T \times S}$; The quantile bins number n_q ;

Output: Anomaly score sets $\mathbf{MA} = \{\mathbf{ma}_1^T, \dots, \mathbf{ma}_S^T\}$, $\mathbf{DA} = \{\mathbf{da}_1^{T/144}, \dots, \mathbf{da}_S^{T/144}\}$ and anomaly points

$\mathbf{anp}_{min} = \{\mathbf{anp}_{min}^1, \dots, \mathbf{anp}_{min}^S\}$, $\mathbf{anp}_{day} = \{\mathbf{anp}_{day}^1, \dots, \mathbf{anp}_{day}^S\}$ for every 10 minutes and every day, respectively;

1: Set the number of quantile bin sets for every 10-minute and daily time intervals as $n(\mathbf{Q}_{min}) = n_q$ and $n(\mathbf{Q}_{day}) = n_q$;

// self-transition probability calculation

2: **for** each column i in $\mathbf{V}_{T \times S}$ **do**

3: Compute the MTF matrix $\mathbf{M}_i^{min}$ and $\mathbf{M}_i^{day}$ of road i using Eq. (4);

4: Assign the main diagonal of $\mathbf{M}_i^{min}$ and $\mathbf{M}_i^{day}$ to the anomaly score vector $\mathbf{ma}_i^T$ and $\mathbf{da}_i^{T/144}$;

5: Store the division result in $\mathbf{ma}_i^T$ and $\mathbf{da}_i^{T/144}$;

6: **end for**

7: Set $\mathbf{MA} = \{\mathbf{ma}_1^T, \dots, \mathbf{ma}_S^T\}$, $\mathbf{DA} = \{\mathbf{da}_1^{T/144}, \dots, \mathbf{da}_S^{T/144}\}$, $\mathbf{Q}_{min} = \{\mathbf{Q}_{min}^1, \dots, \mathbf{Q}_{min}^S\}$, and $\mathbf{Q}_{day} = \{\mathbf{Q}_{day}^1, \dots, \mathbf{Q}_{day}^S\}$;

8: **for** each column j in $\mathbf{S}$ **do**

```

// anomaly detection for the granularity of 10-minute
9:   Choose the topside values of  $\mathbf{Q}_{min}^j$  for every 10 minutes as the threshold of road  $j$ , denoted
      as  $Q_{min}^{j, 1/n_q}$ ;
10:  if  $ma_j^t \geq Q_{min}^{j, 1/n_q} (ma_j^t \in \mathbf{ma}_j^T)$  then
11:    Designate the anomaly point  $anp_{t,min}^j$  (time  $t$ , road  $j$ );
12:  end if
13:    Set the anomaly points (with the total number of  $m$ ) of road  $j$  as
       $\mathbf{anp}_{min}^j = \{anp_{1,min}^j, \dots, anp_{m,min}^j\}$ ;
// anomaly detection for the granularity of one-day
14:  Apply K-means method on the anomaly score set  $\mathbf{DA}$ ;
15:    Designate high anomaly score values as anomaly points cluster
       $\mathbf{anp}_{day}^j = \{anp_{1,day}^j, \dots, anp_{d,day}^j\}$ ;
16: end for
17: Set the anomaly point sets  $\mathbf{anp}_{min} = \{\mathbf{anp}_{min}^1, \dots, \mathbf{anp}_{min}^S\}$  and  $\mathbf{anp}_{day} = \{\mathbf{anp}_{day}^1, \dots, \mathbf{anp}_{day}^S\}$ .

```

As in Algorithm 2, the algorithm starts by dividing the time-series data values into quantile bins. Considering the trade-off between computational efficiency and division accuracy, the number of quantile bins n_q is set within 10~20 for both 10-minute and daily granularity scenarios (line 1 in Algorithm 2). The main diagonals of the MTF matrix, $\mathbf{MA} = \{\mathbf{ma}_1^T, \dots, \mathbf{ma}_S^T\}$ and $\mathbf{DA} = \{\mathbf{da}_1^{T/144}, \dots, \mathbf{da}_S^{T/144}\}$, are computed for each road under two temporal granularities. At the same time, the results of Q bins' division in $\mathbf{MA}$ and $\mathbf{DA}$ are also recorded as $\mathbf{Q}_{min} = \{\mathbf{Q}_{min}^1, \dots, \mathbf{Q}_{min}^S\}$ and $\mathbf{Q}_{day} = \{\mathbf{Q}_{day}^1, \dots, \mathbf{Q}_{day}^S\}$. These two steps, which can be regarded as the self-transition probability calculation, are shown in lines 2~7 in Algorithm 2.

Subsequently, temporal anomaly detection is conducted at different granularities. For temporal anomaly detection with the granularity of every 10 minutes (lines 8~13), the algorithm sets a threshold value for each road based on the topside value $\mathbf{Q}_{min}^j$ of road j . If the self-transition probability exceeds the threshold value, the timestamp is designated as an anomaly point. The algorithm then stores the anomaly points every 10 minutes in $\mathbf{anp}_{min} = \{\mathbf{anp}_{min}^1, \dots, \mathbf{anp}_{min}^S\}$ for all roads.

For temporal anomaly detection with the granularity of every day (lines 14~16), the algorithm applies

the K-means clustering algorithm on the anomaly score set $\mathbf{DA} = \{\mathbf{da}_1^{T/144}, \dots, \mathbf{da}_S^{T/144}\}$. The cluster number is chosen using the elbow method. The algorithm then designates the day in the cluster of high anomaly score values as anomaly points of road j , denoted as $\mathbf{anp}_{day}^j = \{anp_{1,day}^j, \dots, anp_{d,day}^j\}$.

Finally, the algorithm outputs the anomaly sets, $\mathbf{anp}_{min} = \{\mathbf{anp}_{min}^I, \dots, \mathbf{anp}_{min}^S\}$ and $\mathbf{anp}_{day} = \{\mathbf{anp}_{day}^1, \dots, \mathbf{anp}_{day}^S\}$, for every road, under two temporal granularities (line 17). These sets can be used to identify and analyze temporal anomalies in traffic data.

4. Result and Analysis

Based on the above methodology, this section details the experimental design to validate the proposed hybrid data-driven method for spatial and temporal anomaly detection. First, we illustrate the spatial anomaly detection results, identifying the anomalies to showcase their patterns, alongside a quantitative validation to test the spatial anomaly detection method's effectiveness. Subsequently, the effectiveness of the temporal anomaly detection method is evaluated using two distinct datasets. Further analysis of the temporal anomaly detection results is conducted across two distinct temporal granularities. This section confirms the method's ability to capture complex dependencies and dynamic temporal evolutions in urban traffic.

4.1 Data Description

4.1.1 Traffic speed data from three cities

Real-world masking traffic speed data of three cities (Chengdu, Xian, and Shenzhen) from 1st March to 30th June 2018 are used to test the proposed spatial-temporal anomaly detection method. The data contain road ID and corresponding speed records every 10 minutes. A data pre-processing step is performed to remove roads that (i) experienced more than ten percent speed data loss and (ii) are located outside the central area of the city. These criteria exclude roads with poor data quality while retaining the main urban area. The rest of the missing values are filled by the linear interpolation method. After the data pre-processing procedure, the traffic speed data of the three cities are chosen as test bases, which contain every

10-minute traffic speed data (km/h) of 628, 659, and 707 road sections for Chengdu, Xian, and Shenzhen City, respectively. For simplicity, the names of these three cities will be shortened to CD, XA, and SZ.

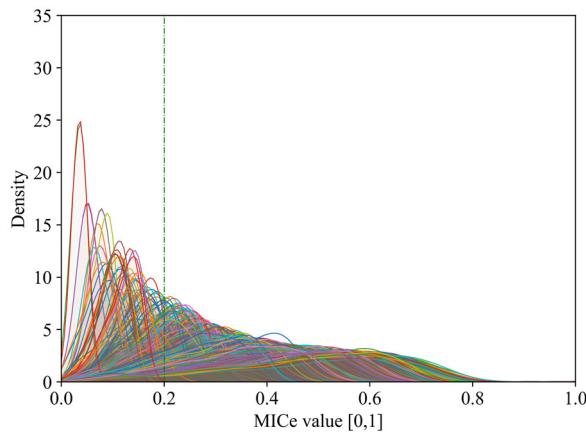
4.1.2 Traffic speed data from PeMS

As anomaly labels are not available in the time series data of the three cities, we faced similar challenges to those described in the former study (Wang and Sun, 2021). To address this, we conducted additional comparison experiments using a different data source provided by PeMS to evaluate the performance of our method in detecting temporal anomalies. The dataset comprises time-series speed data collected every five minutes from a single sensor of I-880 between 8:00 p.m. and 12:00 p.m. from 1st May to 31st July 2018. The dataset was carefully calibrated, containing 188 anomalies in 17,664 time stamps.

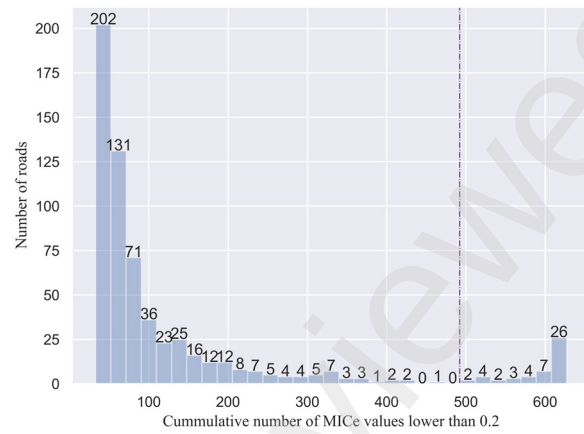
Speed data can be collected from loops, sensors, GPS equipment installed in taxis, and even smartphones of ride-hailing services. Also, speed data is the major data source in most ITS deployed in cities and plays a critical role in identifying traffic conditions on urban roads. By integrating these datasets, this study aims to comprehensively assess the effectiveness and applicability of the proposed spatial and temporal anomaly detection method.

4.2 Spatial Anomaly Detection Considering Spatial Correlations

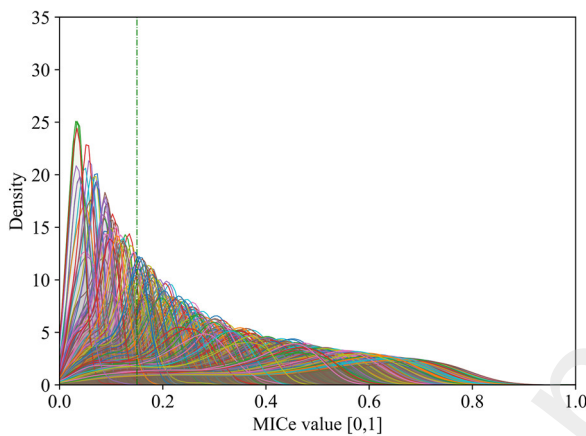
Three spatial dependency matrices of CD, XA, and SZ ($\mathbf{SD}_{628 \times 628}$, $\mathbf{SD}_{659 \times 659}$, and $\mathbf{SD}_{707 \times 707}$) are computed through the speed matrix of each city. Then, these spatial dependency matrices are adopted in the spatial anomaly detection algorithm as described in Algorithm 1. As per Algorithm 1, the initial step involves selecting a threshold value, denoted as p . Intuitively, the threshold can be set according to the distribution of spatial dependence values. Therefore, kernel density estimate (KDE) plots of different cities are illustrated to visualize the frequency distribution (Fd) of spatial dependence values of road sections, as shown in Fig.3.



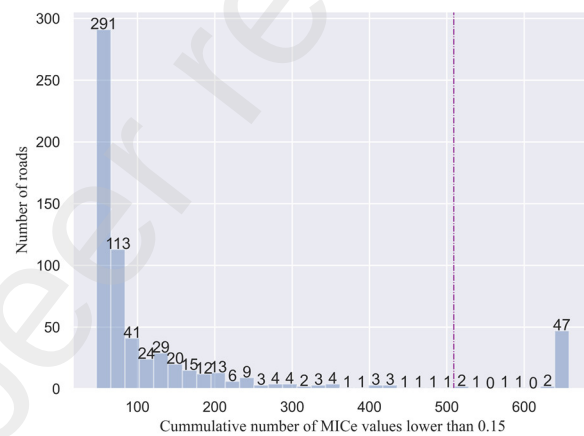
(a) KDE plot of CD



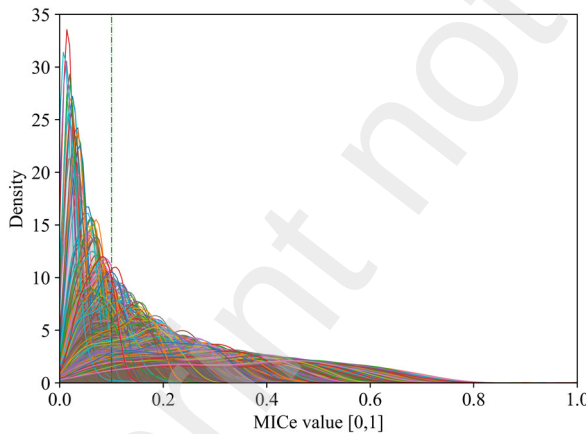
(d) Fd of roads in CD



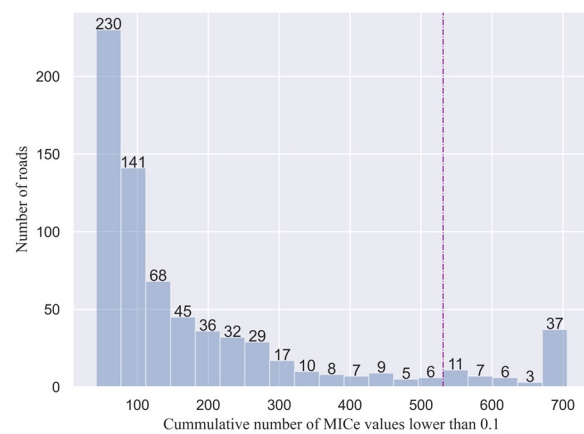
(b) KDE plot of XA



(e) Fd of roads in XA



(c) KDE plot of SZ



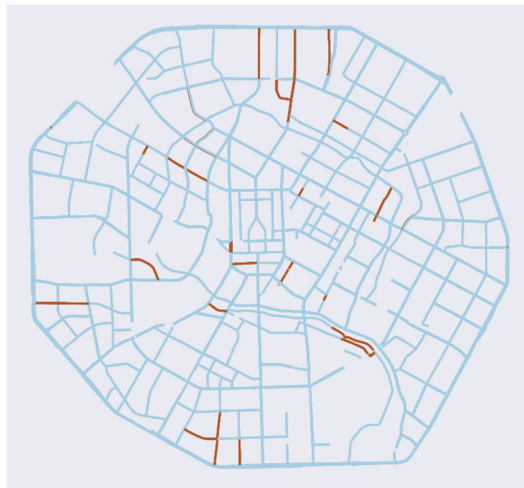
(f) Fd of roads in SZ

Fig. 3. KDE plot of spatial dependence values and frequency distribution of roads.

The top of Fig. 3 displays the KDE plots of the spatial dependency matrices for the three cities. Each KDE line corresponds to a specific road section, distinguished by different colors. The x-axis represents the MIC_e values, and the y-axis is the estimated density of MIC_e values. Notably, certain

road sections in each city exhibit spatial dependencies that cluster around lower levels (indicated by green dot-dash lines), namely below 0.2, 0.15, and 0.1 for CD, XA, and SZ, respectively. It suggests relatively weak correlations between these roads and others. Accordingly, the thresholds for CD, XA, and SZ are set at 0.2, 0.15, and 0.1, respectively. At the bottom of Fig.2, the cumulative numbers of MIC_e values that fall below these thresholds are computed for each road in the different cities, and the resulting frequency distribution is depicted. In Fig.3 (d), (e), and (f), the x-axis denotes the cumulative number of MIC_e values below the threshold for each road, and the y-axis is the number of roads located in a specific range determined by Freedman-Diaconis rules. Each histogram is divided by a purple dot-dash line, the critical boundary.

It is important to note that spatial anomaly detection operates assuming that the frequency distribution of road segments concerning the MIC_e value accumulation follows a long-tail distribution under normal conditions. The presence of a long-tail distribution to the left of the purple dot-dash line represents a widespread phenomenon in complex networks, signifying that a small number of road segments have a large cumulative count below the predetermined threshold. This observation is consistent with spatial correlation patterns, where most roads in the network exert mutual influence upon one another. Roads located on the right side of the purple line deviate from this distribution pattern and are thus identified as anomalies; these roads exhibit a significant quantity of MIC_e values falling below the threshold. For instance, in Fig.3 (d), the rightmost histogram indicates 26 roads with over 600 MIC_e values below 0.2. Note that anomaly detection relies on the assumption that, under normal conditions, the frequency distribution of roads regarding the cumulative number of MIC_e values follows a long-tail distribution. By employing Algorithm 1, the anomaly road sets can be detected, with the numbers 48, 54, and 64 for CD, XA, and SZ, respectively.



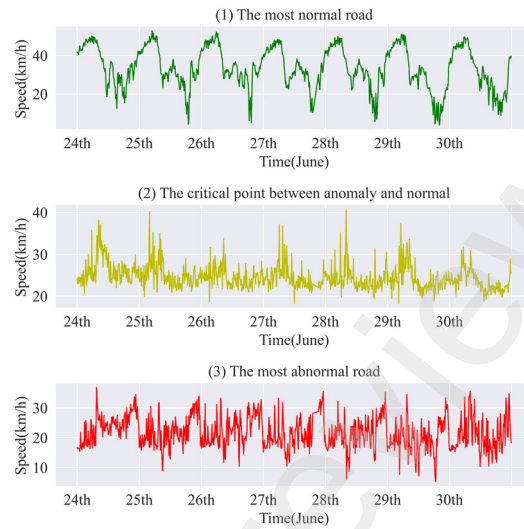
(a) Abnormal roads in CD



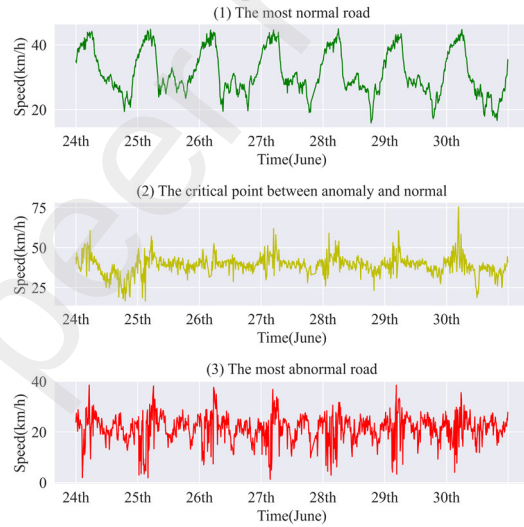
(b) Abnormal roads in XA



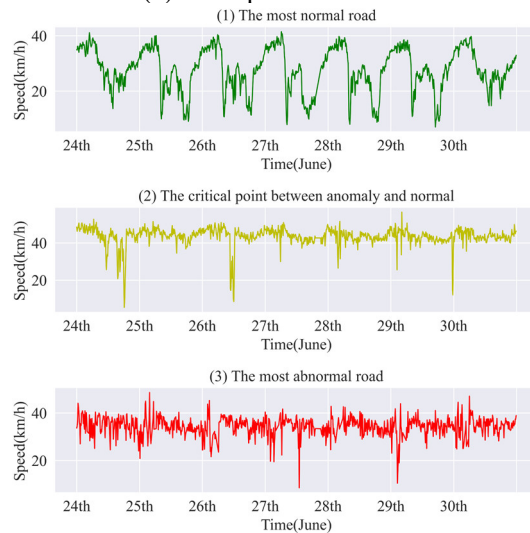
(c) Abnormal roads in SZ



(d) Road patterns in CD



(e) Road patterns in XA



(f) Road patterns in SZ

Fig. 4. Results of the spatial anomaly detection algorithm

Following the stream of Algorithm 1, the results of the spatial anomaly detection algorithm are presented in Fig.4. In Fig.4(a), (b), and (c), the identified spatial anomalies are marked in brown. These anomalies are predominantly found near the railway station, scenic spots, central business district (CBD), beltway, and urban roads along the river. Although some of these anomalies may be attributed to loop failures or measurement errors, it is intriguing to note that these abnormal roads, detected through the analysis of spatial dependencies in time-series data, appear to be potential bottlenecks that warrant additional attention from traffic management authorities.

In Fig.4(d), (e), and (f), three types of typical roads are selected to present the differences between normal and abnormal roads. Specifically, the green line plots depict the most normal roads, characterized by the minimal cumulative number of MICE values that are below the thresholds (as observed in the leftmost histograms of Fig. 3 (d), (e), (f)). These normal roads exhibit typical daily traffic patterns, with distinct morning and evening peak hours clearly evident. The yellow line plots represent the most abnormal roads, showcasing the maximum cumulative number of MICE values below the thresholds (as observed in the rightmost histograms of Fig. 3 (d), (e), (f)). These anomalous roads lack distinct periodic peak hours, displaying chaotic and random characteristics. This could be attributed to their unique geographical location, traffic planning issues, or other factors. Lastly, the red line plots illustrate the critical point roads, which serve as the boundary between normal and abnormal roads (indicated by the purple dot-dash lines in Fig. 3 (d), (e), (f)). These roads demonstrate some degree of daily periodic patterns, yet compared to the most normal roads, their periodic characteristics are less pronounced, indicating lower levels of deterministic behaviors (Vlahogianni et al., 2008).

To further validate our proposed spatial anomaly detection method quantitatively, we employ the *determinism* (DET) metric from Recurrence Quantification Analysis (RQA) (Huang et al.,

2023; Vlahogianni and Karlaftis, 2012). The DET metric measures the determinism of time series data, with values ranging from 0 (100% stochastic, i.e., fully chaos) to 1 (100% deterministic). As the spatial anomalies defined before, the DET metric can then be used to evaluate whether time-series data from a road section is abnormal or not. A larger DET value means that the time-series data of one road is more deterministic with regular periodic patterns, that is, less abnormal in the spatial domain. A validation result of three kinds of typical roads from CD, XA, and SZ, is shown in Fig. 5.

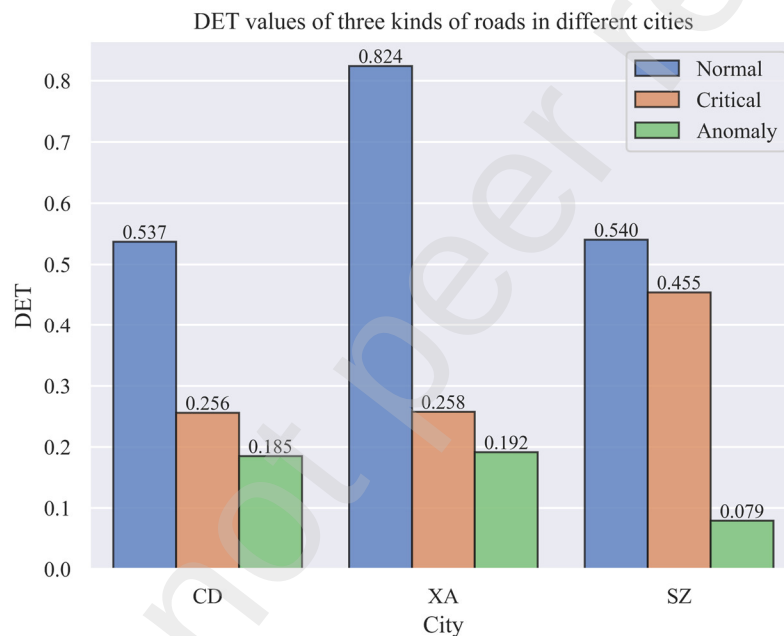


Fig. 5. DET values of three types of roads in different cities

From Fig. 5, it is observed that DET values demonstrate a decreasing trend transitioning from normal roads to abnormal roads. This indicates a gradual intensification of chaotic characteristics in the time-series data as one transitions from normal to anomalous roads. This changing pattern validates the effectiveness of the spatial anomaly detection method in identifying roads with high levels of chaos and anomalous behavior. Additionally, it confirms the accuracy of selecting critical roads based on the long-tail distribution assumption. Notably, in SZ, the critical road shows a smaller DET difference relative to normal roads than in the other two cities. This may be due to a

larger number of roads in the ‘tail’ of the long-tail distribution, as demonstrated in Fig. 3(f).

In summary, these qualitative and quantitative results together affirm the effectiveness of the proposed spatial anomaly detection algorithm, providing a powerful tool for analyzing and detecting anomalies in traffic patterns within urban traffic systems.

4.3 Temporal Anomaly Detection Considering Traffic Evolutions

This subsection focuses on validating the effectiveness of the temporal anomaly detection method and analyzing patterns of the anomaly detection results. First, performance comparisons are conducted between the proposed MTF-based method and other popular anomaly detection methods using well-labeled traffic time-series data. Second, the temporal anomaly detection algorithm is applied to data from three cities. Given that the datasets for the three cities lack actual anomaly labels, this study indirectly evaluates the performance of the method by comparing its results with those of various clustering methods. Finally, pattern analysis for the three cities is conducted at two levels of granularity: minute and daily, utilizing the temporal anomaly detection results from the perspectives of individual roads and the entire network.

4.3.1 Effectiveness validation in well-labeled data

Comparison experiments utilize well-labeled PeMS data, as described in section 4.1.2. This dataset was carefully calibrated, comprising 188 anomalies across 17,664 timestamps. We compared our method with several well-established time-series anomaly detection methods, including KNN (Ramaswamy et al., 2000), LinearPCA (Veeramachaneni et al., 2016), KernelPCA (Hoffmann, 2007), iNNE (isolation using Nearest Neighbor Ensemble) (Bandaragoda et al., 2018), IF (Isolation Forest) (Liu et al., 2012), BD (Boundary and distance) (Jiang et al., 2011), WFRDA (Weighted Fuzzy-Rough Density-based Anomaly) (Yuan et al., 2023). We evaluated the performance of each method using Precision rate, Recall Rate, and AUC metrics, which are

commonly used in highly imbalanced traffic time-series anomaly detection tasks. A higher value for these metrics indicates a better detection performance. The comparison results are presented in

[Table 1](#), and the best performance is bold-marked.

Methods	Precision	Recall	AUC
KNN	75.30%	77.70%	0.887
LinearPCA	38.00%	40.43%	0.699
KernelPCA	79.80%	81.90%	0.908
iNNE	58.46%	60.64%	0.801
IF	71.60%	75.00%	0.873
BD	59.90%	58.00%	0.788
WFRDA	86.80%	87.20%	0.935
MTF(this paper)	85.60%	88.30%	0.941

It can be seen from [Table 1](#) that our method performs the best with the highest recall rate and AUC value, although it has a slightly lower precision rate of 85.6% compared to the WFRDA method, which is based on rough set theory. It is worth noting that for this test dataset with nearly 20,000 timestamps, our MTF-based detection method can be executed in seconds through parallel computing techniques, whilst WFRDA requires almost 1,500 seconds. In summary, our method demonstrates superior performance in temporal traffic anomaly detection tasks.

4.3.2 Effectiveness validation using data from the three cities

As there is a lack of accurate anomaly labels in the time-series data from the cities CD, XA, and SZ, in this subsection, the temporal anomaly detection task is transformed into an unsupervised clustering problem, and the effectiveness of the proposed method is assessed through the evaluation of clustering results. Based on the speed data at ten-minute granularity in the three cities, several clustering methods are employed for comparative analysis, including K-means ([MacQueen, 1967](#)), Gaussian Mixture Model (GMM) ([Reynolds, 2009](#); [Zhang et al., 2021](#)), Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) ([Sathiaraj et al., 2019](#); [Zhang et al., 1996](#)),

and Agglomerative Clustering (Agglomerative) (Murtagh and Contreras, 2012; Pasupathi et al., 2021).

In evaluating clustering results, compactness and separation serve as criteria to group similar objects within the same cluster and ensure they remain distinct across different clusters. Previous studies (Ezugwu et al., 2022; Liu et al., 2010) have shown that the S_Dbw index excels across five key aspects: monotonicity, density, subclusters, skewed distributions, and noise. These aspects are crucial for anomaly detection, notably skewed distributions, and noise, which indicate data with outliers. We selected S_Dbw as an evaluation metric for assessing the results. S_Dbw is calculated as follows:

$$f(x, u) = \begin{cases} 0, & \text{if } d(x, u) > \frac{1}{NC} \sqrt{\sum_{i=1}^{NC} \|\sigma(C_i)\|} \\ 1, & \text{otherwise} \end{cases} \quad (6)$$

$$Scat(NC) = \frac{1}{NC} \sum_{i=1}^{NC} \frac{\|\sigma(C_i)\|}{\|\sigma(D)\|} \quad (7)$$

$$Dens_bw(NC) = \frac{1}{NC(NC-1)} \sum_{i=1}^{NC} \left[\frac{\sum_{x \in C_i \cup C_j} f(x, u_{ij})}{\sum_{j=1, j \neq i}^{NC} \max \left\{ \sum_{x \in C_i \cup C_j} f(x, c_i), \sum_{x \in C_i \cup C_j} f(x, c_j) \right\}} \right] \quad (8)$$

$$S_Dbw(NC) = Scat(NC) + Dens_bw(NC) \quad (9)$$

where, D is data set and NC is the number of clusters; C_i is the i -th cluster and $\sigma(C_i)$ is the variance vector of C_i ; c_i and c_j are centers of cluster C_i and C_j , respectively; u_{ij} is the middle point of the segment defined by cluster centers; $d(x, u)$ represent the distance between x and u ; $\|\cdot\|$ denotes the l2-norm calculation; $Scat(NC)$ and $Dens_bw(NC)$ indicates compactness and separation, respectively; $S_Dbw(NC)$ is S_Dbw value of the cluster's result and a small value of $S_Dbw(NC)$ indicates well-separated clusters.

Our method, designated as MTF, was tested alongside the four previously mentioned methods

on each road across three cities. The number of clusters for all methods was aligned with that of the MTF method to ensure a fair comparison. Table 2 displays the mean values and standard deviations of the S_Dbw index of the whole network. The best results, characterized by the lowest mean values and standard deviations for each city, are highlighted in bold.

Table 2 S_Dbw result of different methods

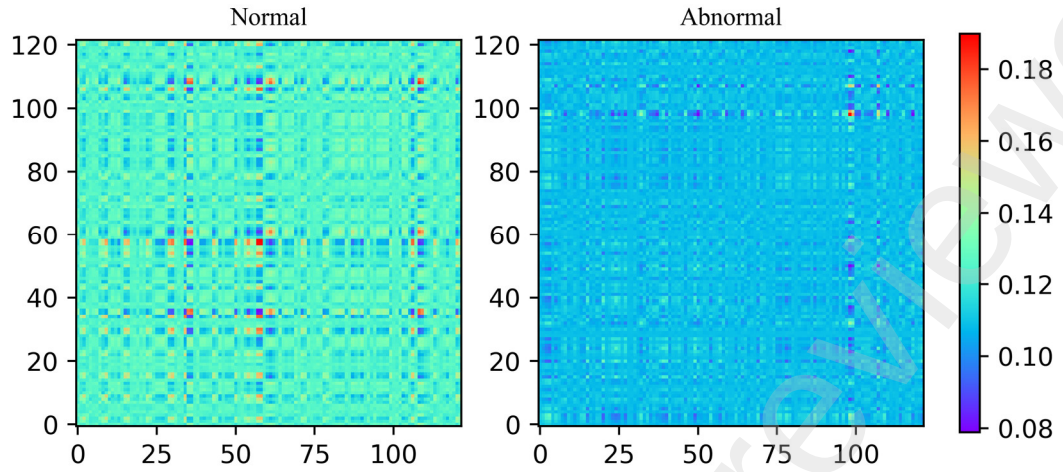
Method	Mean value			Standard deviation		
	CD	XA	SZ	CD	XA	SZ
K-means	0.407	0.420	0.442	0.047	0.064	0.058
GMM	0.409	0.422	0.446	0.052	0.066	0.068
BIRCH	0.403	0.420	0.436	0.056	0.075	0.065
Agglomerative	0.408	0.421	0.441	0.047	0.063	0.056
MTF(this paper)	0.408	0.417	0.431	0.043	0.051	0.044

The MTF method clearly exhibits superior performance, achieving almost the lowest mean values (except in CD city) and the lowest standard deviation values for the S_Dbw index across three cities. The results showcase not only the efficiency and precision of our method during the clustering process, characterized by notable compactness and separation of clusters but also its reliability across varied road conditions within a road network. This underscores the proposed method's high adaptability and robustness in managing complex urban traffic systems and offers reliable evidence supporting its application in various cities in the future.

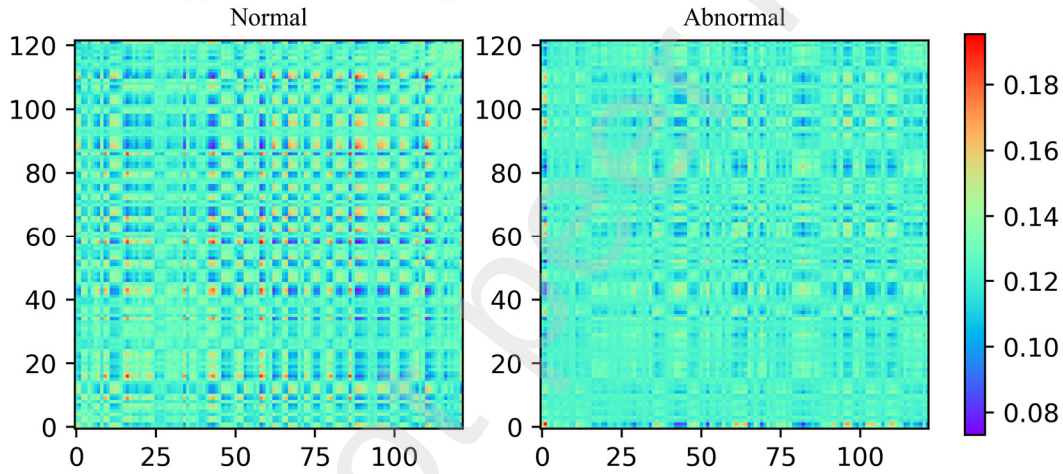
4.3.3 Pattern analysis of MTF matrix

In Algorithm 2, the MTF matrix serves a fundamental role by providing traffic transition probabilities in time-series data, which can further be utilized to quantify traffic evolution on roads. This subsection will present illustrations and examine the distinct patterns of the MTF matrix for different types of roads across three cities. Fig. 6 displays the MTF matrices used in the temporal anomaly detection algorithm, showcasing heatmaps of the MTF matrices for the most normal and abnormal roads in each of the three cities. A sample time of 24 hours (one day) is selected to ensure

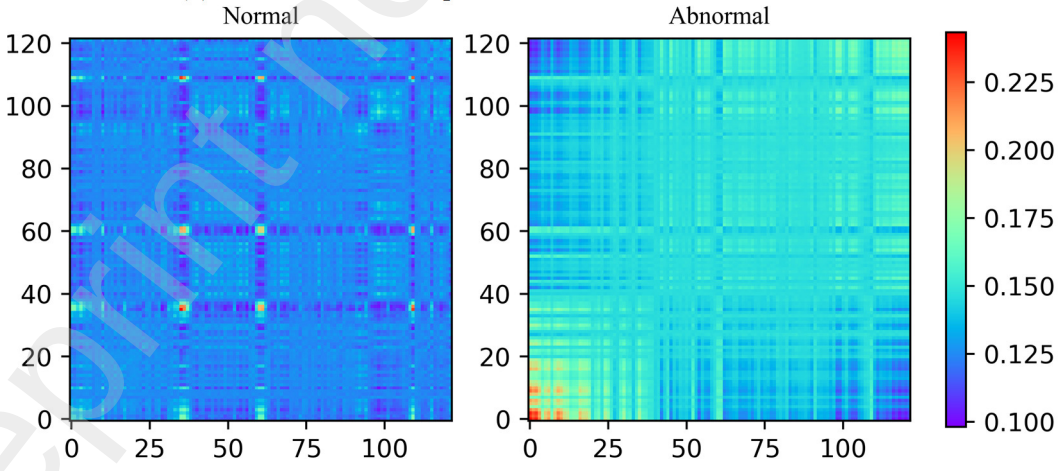
clarity in the presentation.



(a) MTF matrix heatmap for normal and abnormal roads in CD



(b) MTF matrix heatmap for normal and abnormal roads in XA



(c) MTF matrix heatmap for normal and abnormal roads in SZ

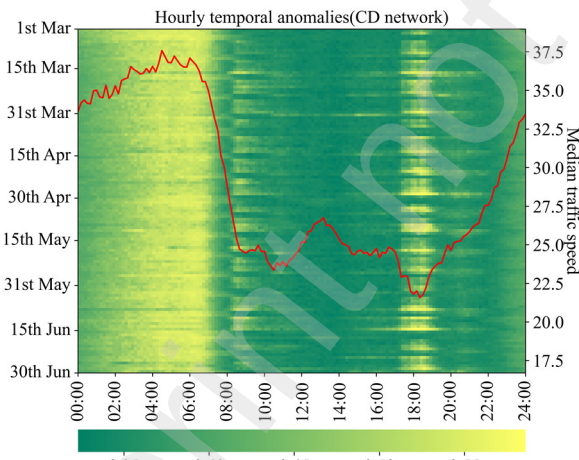
Fig. 6. Illustration of MTF matrices of the normal roads and abnormal roads in CD, XA, and SZ

The color bars on the right side of Fig.6(a), (b), and (c) denote the transition probabilities of

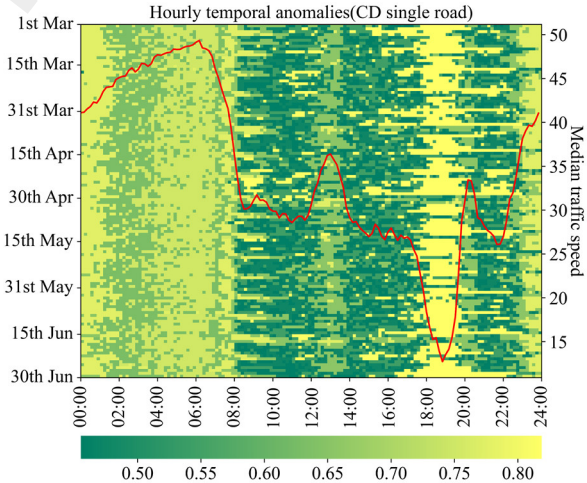
elements in the MTF matrix. On the left side of Fig.6(a), (b), and (c), a clear periodic characteristic is observed in the MTF matrix of the normal road in each city. This periodic pattern is manifested both in the main diagonal and across the entire matrix, reflecting the inherent property of daily traffic under normal conditions. Notably, the comparison between the normal and abnormal roads in each city reveals significant differences. In the MTF matrix of each city, the latticed blocks corresponding to the normal road are much more prominent and evenly distributed. This finding serves as additional supporting evidence for the effectiveness of our spatial anomaly detection algorithm, as it highlights the lack of apparent periodicity in the daily traffic of abnormal roads.

4.3.4 Temporal anomaly detection: a minute granularity level

As described in Algorithm 2, the main diagonal (self-transition probability) of the MTF matrix is selected for detecting temporal anomalies. After the temporal anomaly detection algorithm, we present **MA**, the anomaly score every 10 minutes under different scenes (the most normal single road and network scale) in Fig. 7.



(a) Network scale self-transition and median speed values of CD



(d) Self-transition and median speed values of a single road in CD

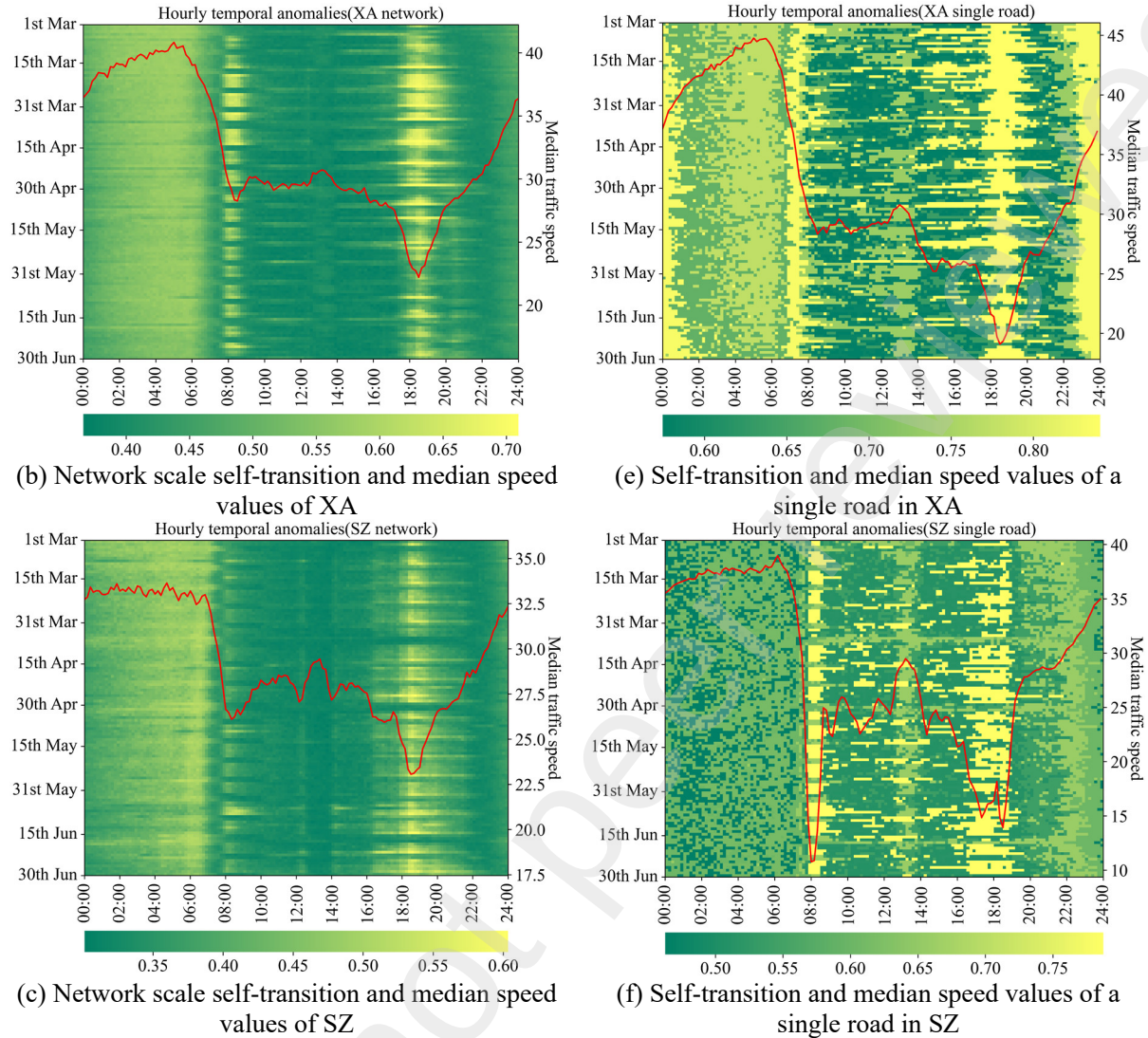


Fig. 7. Self-transition probabilities and median speed values every 10 minutes

In all sub-plots of the three cities, the x-axis is the time of day, and the y-axis on the left side is the date. Further, a line plot of the median traffic speed every 10 minutes (marked in red) on different dates is added on the right-side y-axis to explore the relationship between daily traffic speeds and anomaly scores. The color bar on the bottom of each subplot is decided by its own anomaly score set **MA** under different scenarios.

Fig. 7 depicts the distribution of anomaly scores in both the network scale (Fig. 7(a), (b), and (c)) and individual roads (Fig. 7(d), (e), and (f)). Despite some variations resulting from averaging operations across roads, a similar pattern of anomaly score distribution can be observed. High

anomaly scores (indicated by bright yellow areas) are concentrated during specific time periods, including midnight (from 3 a.m. to 7 a.m.), morning rush hours (from 8 a.m. to 9 a.m.), and evening rush hours (from 5 p.m. to 7 p.m.). It is not surprising that the high anomaly scores gather in these periods because traffic speed data in the urban network remain at a pretty high (midnight) or low (rush hours) level, which increases the probability of maintaining their current states. During rush hours, the congestion slows down traffic propagation along the road, leading to low inter-transition probabilities to others and higher self-transition probabilities. Similarly, during midnights, the urban traffic is in free flow conditions with relatively high speed, allowing roads to maintain their current states.

Furthermore, a detailed illustration of the temporal anomaly detection results for every 10 minutes (i.e., the temporal anomaly points $\mathbf{anp}^{min}$ on a single road) is shown in Fig.8. The snapshot in Fig.8 illustrates the detection result for the most normal road in CD during the period from 8th May to 11th May. The reason for choosing this period is that two typical types of temporal anomalies are included in this period through manual verifications. It is a suitable example to demonstrate the effectiveness of the proposed algorithm.

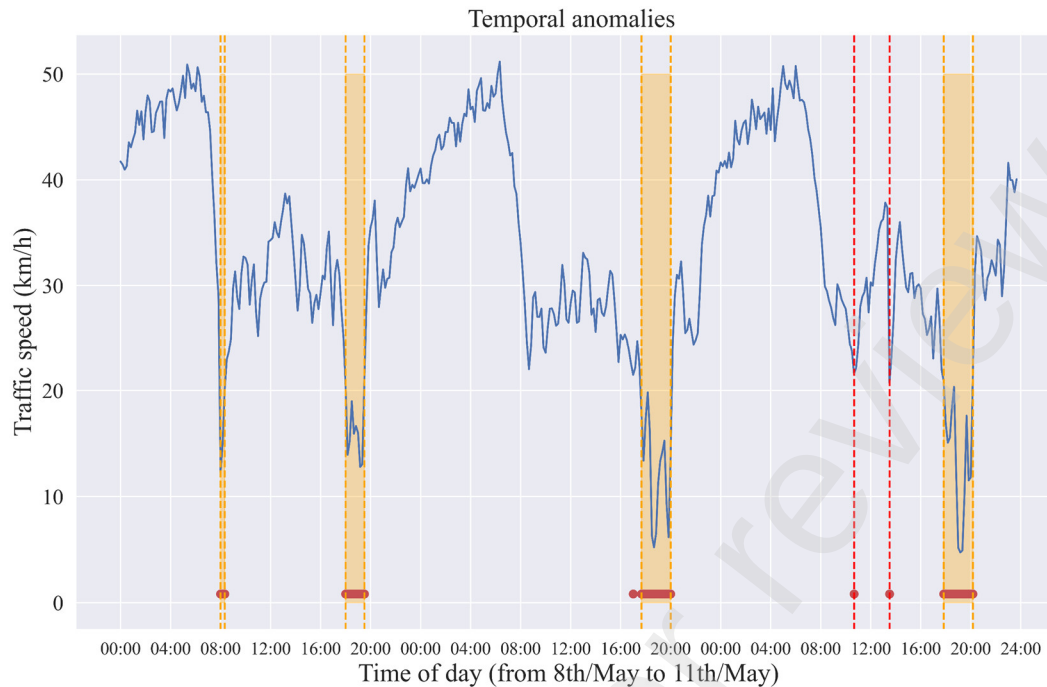


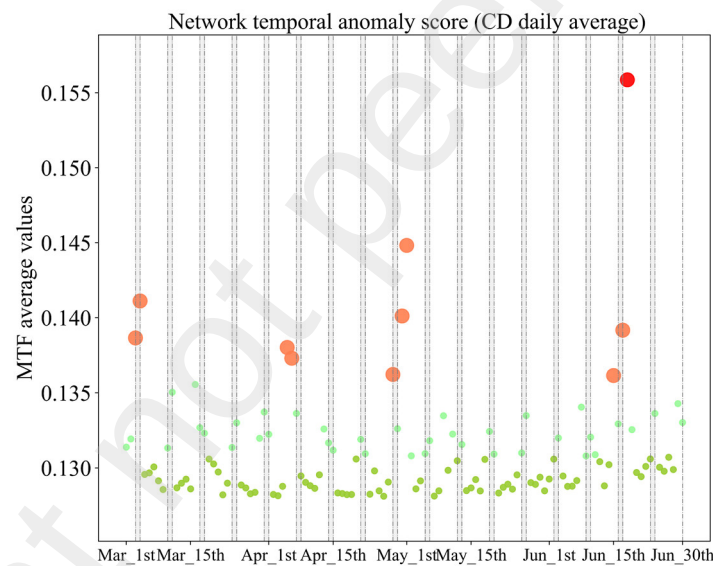
Fig. 8. Fragments of temporal anomaly detection results for a single road in CD.

In Fig.8, the x-axis represents the time of day, and the y-axis is the traffic speed. The red points correspond to detected anomalies, which can be categorized into point anomalies (marked by red dashes) and period anomalies (marked by yellow dashes). It is clear that anomalies occur during periods of rapid change and valleys in the time series. In the case of period anomalies, the start and endpoints are denoted by yellow dash lines and are filled in yellow. Typically, period anomalies arise during rush hours. For example, most period anomalies in the chosen road are concentrated between 6 p.m. and 8 p.m., corresponding to the evening rush hour. Also, from the line plots of traffic speed Fig. 7d), we can observe that the duration of the morning rush hour on the chosen road is shorter compared to that of the evening rush hour. This explains why only one morning rush hour (around 8 a.m. on 8th May) is detected in Fig.8. Regarding point anomalies, three anomalies are detected and marked by red dashed lines. One anomaly is located near the rush hours on 9th May, and we speculate that it can be considered part of an extended period of high traffic congestion. The other two anomalies occur around 12 p.m., during which sudden drops in traffic

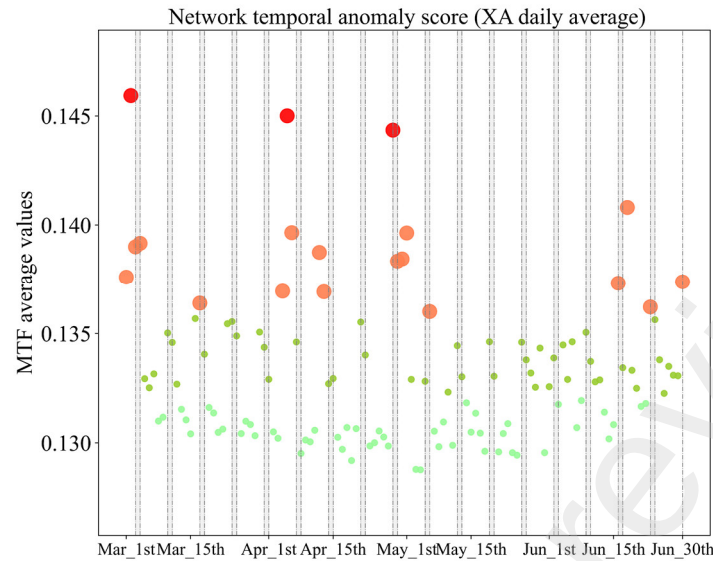
speed occur, likely caused by accidents, monitoring faults, or other unforeseen factors.

4.3.5 Temporal anomaly detection: a daily granularity level

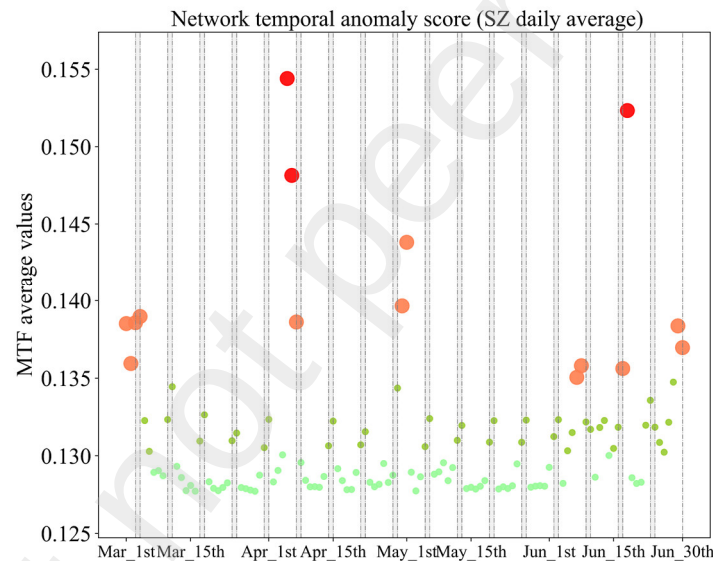
To analyze the performance of the temporal anomaly detection algorithm under daily granularity, we aggregate the anomaly score set of different roads in each city to the network scale. This is done by calculating the average MTF values for each road. The results are presented in Fig.9, where the daily temporal anomalies $\mathbf{anp}^{day}$ are identified according to Algorithm 2, and the vertical gray dot-dash lines label weekends. The scatter plot displays different colors to represent the cluster results obtained from the k-means algorithm. The number of clusters is determined as 4 using the elbow method.



(a) CD, March-June



(b) XA, March-June



(c) SZ, March-June

Fig. 9. Network temporal anomaly scores from 1st March to 30th June.

In Fig.9(a), (b), and (c), the MTF values of the network are grouped into four clusters, where two clusters with the highest values (colored in red and coral) exhibit distinct patterns compared to the other clusters. These two groups are identified as temporal anomaly sets for daily granularity. Upon examining the dates associated with these groups, it becomes evident that they correspond to holidays observed in China. Additionally, other instances related to public activities specific to each city are not listed here. This observation indicates that the temporal anomaly detection

algorithm can identify holidays and public activities that significantly impact urban traffic patterns.

Another interesting finding is the presence of a weekly periodicity in the anomaly scores (MTF values). In Fig.9, the weekends are indicated by vertical gray dot-dash lines. Apart from the temporal anomaly sets and the days directly affected by them, the anomaly scores exhibit a consistent pattern. From Monday to Thursday, the scores remain relatively low, but they experience a significant increase from Friday to Sunday. This pattern aligns with the daily traffic patterns observed in urban cities, where traffic flows on weekends and Fridays differ from those on weekdays, and thus can be considered as anomalies.

In summary, the temporal anomaly detection algorithm can detect anomalies from different temporal granularities. In this paper, we conducted temporal anomaly detections at two specific temporal granularities: every ten minutes and daily intervals. It is worth noting that the proposed method is effectiveness for any other temporal granularities and can be readily adjusted to accommodate new data with different sample time values, making it adaptable and flexible for various time resolutions.

4.4. Discussion

There are still some aspects that require further discussion. First, our objective is to develop an anomaly detection method that can fully take advantage of the profits of data explosion in the new information era. However, transitioning to a mature smart city system is time-consuming, and current systems frequently grapple with limited data environments. Currently, traffic data sources are still limited, and the growing volume of traffic time-series data, being the primary accessible data source, presents a challenge in effectively labeling anomalies for traffic management. As it is, the advantage of this approach lies in its capacity to capture dynamic changes in road speed and abnormal traffic patterns, utilizing only time-series data from individual road segments. This

renders it an efficient data-driven method with minimal operational requirements, accounting for spatial correlations and temporal evolutions in traffic based on speed data.

It is important to acknowledge that integrating temporal and spatial factors can enhance the accuracy of anomaly detection methods. However, some studies in short-term prediction ([Wang et al., 2020](#); [Zhang et al., 2019](#)) have shown that the most significant contributions to prediction results come from the data of the road segment (or detector) itself. The influence of upstream and downstream road segments on the target segment diminishes sharply with increasing distance.¹ Additionally, the minimum sample interval for data in our study is set as 10 minutes. Given the relatively large time interval, the impacts of traffic propagation on nearby road segments may be underestimated, potentially diminishing the advantages of integrating upstream and downstream data in anomaly detection. Considering these factors, we conducted temporal anomaly detection based solely on the speed data of the road segment itself (and, in fact, the results also demonstrate the effectiveness of our method).

Second, the dynamic homogeneity of traffic networks ([Geroliminis and Daganzo, 2008](#)) may affect the spatial anomaly detection algorithm. Homogeneous clusters of roads may exhibit higher values of spatial dependence, causing the KDE of spatial dependence values to aggregate at a low value, and impeding the spatial anomaly detection results. This assumption may weaken the applicability of the spatial anomaly detection algorithm, particularly in cities with many clusters, such as those located in areas with several rivers that naturally group a number of roads into clusters. Therefore, future research should investigate the relationship between dynamic homogeneity and spatial dependence within traffic networks. Adjusting and optimizing the spatial anomaly detection algorithm to improve its applicability in these scenarios requires further

¹ this discussion does not consider the traffic interactions of road segments due to their inherent characteristics (such as Points of Interest and land use attributes)

investigation despite the rarity of such city types compared to typical urban layouts.

Third, although the anomaly detection method has been proven to be effective, some differences exist when applying the method to different cities. Initially, the density of road networks can impact the result of our temporal anomaly detection method. As seen in Fig. 7(b), XA exhibits higher anomaly scores (represented by brighter yellow colors) during morning and evening peaks compared to the other two cities. The road network density in XA is 5.49 km/km^2 , which is lower than CD (8.02 km/km^2) and SZ (9.50 km/km^2). A small road network density restricts the capability of urban networks, resulting in higher anomaly scores. Similar findings are observed in Fig. 9(b), where network anomaly scores of XA are higher than the other two cities. Fortunately, the impacts of such differences are blunted when dealing with the same city.

5. Conclusion

Spatial-temporal anomalies in urban traffic can significantly affect the efficiency of ITS in smart cities, resulting in extra fuel consumption and increased vehicle emissions. This study addresses the challenge of accurately and effectively detecting anomalies in raw traffic data under limited data environments without relying on complex parameter tuning. The proposed data-driven spatial-temporal anomaly detection method offers an efficient solution for handling the dynamic and complex nature of traffic data. Experimental analyses using real-world data validate the effectiveness of the proposed method in identifying abnormal roads with distinct traffic patterns and labeling anomalous periods.

The primary objective of this study is to identify roads that require additional management and control measures while providing a strong foundation for the implementation of potent learning-based anomaly detection methods. The practical implications of this study extend to urban traffic and city management departments. The spatial anomaly detection algorithm enables the

identification of roads with more stochastic traffic patterns that require specialized traffic management measures during daily operations. Furthermore, the temporal anomaly detection algorithm allows for exploring traffic patterns at different time granularities, such as minute-by-minute, hourly, and daily intervals. It can identify sudden speed drops, congestion during rush hours, as well as special events and holidays reflected in speed data. City transportation managers can leverage this information to improve traffic flow and optimize traffic management strategies. By adopting this data-driven approach, the reliance on manual operations can be reduced, and well-labeled data can be generated using basic traffic data, facilitating further development goals.

Future works will focus on developing real-time anomaly detection methods based on this work and deep learning models. Developing methods based on Generative Adversarial Networks (Tang et al., 2023) or tensor decomposition (Xing et al., 2023) to utilize the limited data source will be a promising direction. Additionally, the performance of spatial detection methods on different road types will also be the next research topic.

Acknowledgments

This work is supported by the China Scholarship Council (No. 202207000072), National Natural Science Foundation of China (No. 62103292), and Chengdu Technology Innovation R&D Project of Key R&D Support Program (No. 2022-YF05-00302-SN).

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